

Step 1. Initial Setup:

- Set the supply voltage to **2.7V** using the power supply.
- Verify the exact input voltage with a multimeter before proceeding.
- Note: Record the exact multimeter reading in a data table; this value will serve as the reference.

Step 2. 4-bit SAR ADC Conversion:

- Run the 4-bit SAR ADC conversion.
- At each of the 4 conversion steps, record the ADC output value, the multimeter voltage, and observe the PWM signal on the oscilloscope.
- At the final step, capture a screenshot of the PWM signal from the oscilloscope. Note the **duty cycle** and **mean voltage** readings for Step 4.

Step 3. 10-bit SAR ADC Conversion:

- Repeat the process for the 10-bit SAR ADC, stepping through all 10 conversion steps.
- Record the ADC output value and the multimeter voltage at each step.
- At the final step, capture a screenshot of the PWM signal and note the **duty cycle** and **mean voltage** readings for Step 4.

Step 4. Duty Cycle & Mean Voltage Verification:

- Using the oscilloscope screenshots and readings captured at the final steps, measure the duty cycle (D).
- Calculate the expected mean voltage using the formula:

$$V_{\text{mean}} = D \times V_{\text{supply}}$$

- Compare this calculated value with the oscilloscope's mean voltage reading to verify the relationship.

Step 5. Accuracy Evaluation:

- Compare the final estimated voltage outputs of both the 4-bit and 10-bit ADCs against the reference multimeter reading recorded in Step 1 to evaluate accuracy.

Step 6. Data Visualization:

- Plot the successive approximation progression for both the 4-bit and 10-bit cases.
- Illustrate how the estimated voltage converges toward the true input voltage with each step.

1.2 Total Unadjusted Error (TUE) Calculation

Using the data collected in Section 1.2, calculate the Total Unadjusted Error (TUE) for each measurement point. Compare your measured TUE values with the theoretical Quantization Error calculated previously. Discuss whether the observed errors in the 4-bit mode are consistent with theoretical expectations. Explain the concept of Quantization Error and how bit-depth influences the accuracy of the system.